

Rohith Kumar

GenAI Platform Engineer | LLMOps | RAG | Vertex AI

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PROFESSIONAL SUMMARY

- Over 9+ years of experience in architecting and operationalizing end-to-end **AI/ML/GenAI systems** across cloud-native and hybrid enterprise environments, with expertise in **large language models** (LLMs), retrieval-augmented generation (RAG), vector database integration, scalable MLOps workflows, and low-latency inference infrastructure.
- Proficient in designing modular pipelines for training, fine-tuning, and deploying models, and in managing GPU- workloads, secure API endpoints, and reproducible experimentation environments. Fine-tuned foundation models such as **LLaMA2**, **Mistral**, and **Whisper** using Hugging Face Transformers and deployed inference on **GKE** with **KServe** and **Cloud Run**.
- Strong expertise in **Python**, **SQL**, and **R**, with advanced modeling using **PyTorch**, **TensorFlow**, **scikit-learn**, **XGBoost**, and **LightGBM** across regression, classification, NLP, and time-series tasks. Built Retrieval-Augmented Generation (RAG) systems using **LangChain**, **Vertex AI**, **PaLM API**, and vector stores like **FAISS**, **Pinecone**, **Weaviate**, and **Qdrant**.
- Built multimodal **GenAI microservices** using **CLIP**, **Stable Diffusion**, and **Diffusers** to support production-grade **image and audio understanding** via containerized inference. Adept at bridging data science and DevOps & mentoring teams in GenAI systems design, vector indexing, RAG pipelines, and real-time inference deployment.
- Designed and orchestrated **ML pipelines** with **Airflow**, **TFX**, and **Argo Workflows**, integrating CI/CD using **GitHub Actions**, **Terraform**, and **Jenkins** for automated retraining and deployment.
- Streamlined model lifecycle management with **MLflow**, **DVC**, and **BentoML**, enabling reproducible experiments, artifact versioning, and consistent deployment workflows.
- Deployed high-throughput inference services using **KServe**, **Triton**, **TorchServe**, and **TensorFlow Serving**, exposing models over scalable **REST/gRPC** endpoints.
- Delivered performant vectorized retrieval systems using **FAISS**, **MongoDB**, and **sentence-transformer embeddings**, enabling semantic search and contextual augmentation in RAG systems.
- Designed and evaluated model outputs using metrics such as **BLEU**, **ROUGE**, **perplexity**, and **SHAP**, supporting prompt quality analysis and LLM response grounding validation.
- Managed complex IaC deployments using **Terraform** and **Helm**, codifying cloud-native ML environments.
- Designed scalable, production-grade MLOps workflows on **Kubernetes** (GKE/AKS), with secure secrets handling via **Vault**, **Azure Key Vault**, and **Secret Manager**.
- Provisioned and executed distributed training workflows using **GPU-accelerated compute** on **Vertex AI**, **Azure ML**, and **AWS SageMaker**, leveraging **auto-scaling clusters**, **TPUs**, and **custom container images** to optimize large-model fine-tuning pipelines for **LLaMA2** and **Mistral**.
- Integrated model monitoring and drift detection using **Evidently AI**, **Prometheus**, and **Grafana**, with custom **dashboards and alerts** via **Cloud Monitoring** and **Azure Monitor** to detect data skew and degradation in LLM response quality.
- Built scalable, versioned data ingestion pipelines with **Apache Airflow** and **Cloud Functions**, transforming and syncing feature sets to **Feast**, **BigQuery**, **AWS S3**, and **Azure Blob Storage**, while maintaining **lineage tracking** and **schema consistency** across multiple environments.
- Secured all compute, data, and service access via **OAuth2**, **Cloud IAM**, **RBAC**, and **Terraform-managed policy configurations**.
- Leveraged **PromptLayer** and **Vertex AI Workbench** to monitor prompt execution, debug hallucinations, and refine LLM outputs across **PaLM**, **LLaMA2**, and **Mistral** models.
- Followed GitOps and test-first design principles with **PyTest**, **pre-commit**, and **Flake8**, ensuring ML codebase quality, versioning, and CI test coverage.

TECHNICAL SKILLS

1. **Programming Languages & Scripting:** Python, R, SQL
2. **ML & DL Frameworks:** PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, Statsmodels
3. **LLMs, GenAI, Prompt Engineering & Foundation Models:** LangChain, PaLM API, LLaMA2, Mistral, PromptLayer, CLIP, Whisper, Diffusers, Stable Diffusion
4. **ML Experimentation, Tracking & Pipelines:** MLflow, DVC, TFX, Feast, Apache Airflow, Argo Workflows, BentoML
5. **Model Serving & Inference:** TorchServe, TensorFlow Serving, Triton Inference Server, Flask, Azure Functions, Cloud Run
6. **Vector Stores & RAG Infrastructure:** FAISS, Pinecone, Weaviate, Qdrant, Elasticsearch, Milvus
7. **Monitoring, Drift Detection & Observability:** Evidently AI, Prometheus, Grafana, Azure Monitor, Cloud Monitoring
8. **Data Engineering & Storage:** BigQuery, Azure Blob Storage, AWS S3, Google Cloud Storage, PostgreSQL, MySQL
9. **Containerization & Orchestration:** Docker, Kubernetes, GKE, AKS, Helm

10. **MLOps, CI/CD & Automation:** GitHub Actions, Jenkins, Terraform, Git, Cloud Scheduler
11. **Security, Identity & Access Management:** OAuth2, Vault, Azure Key Vault, Secret Manager, Cloud IAM
12. **Cloud Platforms & Services:** Google Cloud Platform (GCP), AWS, Azure
13. **IDEs, Tools & Environments:** Jupyter Notebook, Vertex AI Workbench, VS Code, Ubuntu, WSL

WORK EXPERIENCE

Predactica – Plano, Texas, US | **GenAI Engineer**

April 2025 – Present

Project: Enterprise Knowledge Retrieval and Generative Intelligence Platform

Scope: Designed and deployed a GenAI-powered RAG platform to deliver enterprise-grade conversational AI and multimodal insights across structured, semi-structured, and unstructured data sources.

- Built Retrieval-Augmented Generation (RAG) pipelines using **LangChain**, **Vertex AI**, and **FAISS**, enabling scalable, real-time document retrieval and synthesis from enterprise knowledge bases.
- Fine-tuned **LLaMA2** and **Mistral** on domain-specific corpora using **Hugging Face Transformers** and deployed inference endpoints on **Vertex AI**, optimized for latency and compliance.
- Integrated **PaLM API** with **LangChain** agents to orchestrate tool-augmented reasoning for multi-step query resolution, with structured logging via **PromptLayer** for prompt engineering evaluation.
- Deployed vector search across hybrid stores (**Pinecone**, **Weaviate**, **Qdrant**) using **Terraform** and **Kubernetes**, enabling dense embedding retrieval with fallback to **Elasticsearch** for keyword-based queries.
- Containerized multimodal pipelines using **CLIP** and **Whisper** via **Docker** and served them using **KServe** on **GKE**, supporting use cases like audio tagging and visual content summarization.
- Integrated **Stable Diffusion** and **Diffusers** into **Cloud Run** microservices, enabling low-latency generation of marketing creatives via secure, token-gated APIs.
- Automated ML pipeline orchestration using **Apache Airflow** and **TFX**, managing ingestion, validation, fine-tuning, and deployment workflows across training and staging environments.
- Tracked model metadata, lineage, and artifacts using **MLflow** and **DVC**, ensuring reproducibility and compliance in experimentation workflows.
- Packaged and deployed GenAI endpoints with **BentoML** and exposed them via **Cloud Functions**, enabling decoupled, event-driven GenAI use cases across the platform.
- Built real-time drift detection and model monitoring dashboards using **Evidently AI**, **Prometheus**, and **Grafana**, pushing alerts to **Cloud Monitoring** with SLA thresholds.
- Integrated **BigQuery** as a feature source and logging sink for conversational analytics, allowing LLM behavioral tracing and performance metrics at scale.
- Used **Vertex AI Workbench** for prompt experimentation, embedding visualizations, and testing multi-modal model interactions across **CLIP** and **Whisper** backends.
- Managed secrets and fine-tuning configs using **Secret Manager**, with role-based access controls enforced via **Cloud IAM** and automated policy checks in CI via **Terraform**.
- Implemented GitOps-driven CI/CD pipelines using **GitHub Actions**, automatically triggering test, container build, and deploy steps on commit to main for ML services on **GKE**.
- Scheduled nightly retraining and prompt audits using **Cloud Scheduler** to trigger Airflow DAGs for usage recalibration, abuse mitigation, and prompt-embedding freshness.

Environment: Python, LangChain, Vertex AI, PaLM API, Hugging Face Transformers, LLaMA2, Mistral, PromptLayer, FAISS, Pinecone, Weaviate, Qdrant, Elasticsearch, Milvus, CLIP, Whisper, Diffusers, Stable Diffusion, KServe, TFX, Ray, Apache Airflow, MLflow, DVC, BentoML, Evidently AI, Prometheus, Grafana, GKE, Cloud Functions, BigQuery, Vertex AI Workbench, Cloud Run, Cloud Scheduler, Cloud Monitoring, Secret Manager, Cloud IAM, Terraform, Kubernetes, Helm

Atlassian – San Francisco, CA | **Senior ML Engineer**

Oct 2023 – Mar 2025

- Developed and productionized distributed training pipelines using **PyTorch** and **TensorFlow** for internal AI assistants, optimizing compute-intensive workflows on **Azure Kubernetes Service (AKS)** with GPU scheduling and autoscaling.
- Built modular, CI/CD-integrated training pipelines using **MLflow**, **DVC**, and **GitHub Actions**, ensuring reproducibility, lineage tracking, and model version governance across multiple projects.

- Integrated **TensorFlow Serving**, **TorchServe**, and **Triton Inference Server** to enable scalable model inference APIs on **AKS**, reducing latency by 45% for real-time collaboration features.
- Orchestrated feature engineering and online/offline feature stores using **Feast** backed by **Azure Blob Storage**, enabling consistent and low-latency features for fraud detection models.
- Automated retraining and deployment pipelines using **Apache Airflow** and **Argo Workflows**, incorporating custom KubernetesPodOperators and MLflow lifecycle events.
- Secured access to deployed models and data pipelines using **OAuth2**, **Azure Key Vault**, and **Vault**, ensuring compliance with internal data protection policies and SOC2 audit requirements.
- Applied **SHAP** and **LIME** for interpretable ML, integrating explainability dashboards into internal tools to support regulatory review for models used in enterprise pricing engines.
- Deployed serverless scoring endpoints using **Azure Functions** that consumed containerized models from **AKS**, auto-scaling with event triggers and routing via Azure API Management.
- Designed infrastructure-as-code deployments with **Terraform** and **Helm** to spin up reproducible ML environments, codifying compute, secrets, storage, and monitoring components on Azure.
- Monitored production model health with **Azure Monitor**, integrating drift detection alerts with **Airflow DAGs** and **Slack** notifications for proactive mitigation.
- Created modular training workflows using **Python**, with loosely coupled tasks abstracted into reusable operators that interact with **MLflow**, **Feast**, and **AKS**, enhancing reusability across teams.
- Streamlined experimentation workflows by integrating **MLflow Tracking**, **GitHub Actions**, and **Azure Blob Storage**, standardizing experiment capture and artifact management across pods.
- Enabled multi-model serving and A/B testing strategies on **Triton Inference Server**, integrating traffic-splitting logic into Kubernetes ingress to evaluate performance in real time.
- Led ML platform hardening by integrating audit logging, secret rotation, and IAM-controlled access to storage and compute across **AKS**, **Azure Key Vault**, and **Vault**.
- Mentored junior MLEs and platform engineers on production-grade model deployment patterns, container orchestration on **AKS**, and principles of scalable, cloud-native MLOps on Azure.

Environment: Python, PyTorch, TensorFlow, MLflow, DVC, Apache Airflow, Feast, Kubernetes, Docker, Helm, Terraform, TensorFlow Serving, TorchServe, Triton Inference Server, Argo Workflows, Azure Machine Learning, Azure Kubernetes Service (AKS), Azure Blob Storage, Azure Functions, Azure Key Vault, Azure Monitor, SHAP, LIME, OAuth2, Vault, Git, GitHub Actions

IBM | Armonk, New York, US | ML Platform Engineer

June 2021 – Sept 2023

Project Scope – Designed and maintained a scalable ML platform supporting experimentation, model versioning, reproducible pipelines, and automated deployment across environments. Focused on compliance-driven ML lifecycle automation and integration with enterprise-grade CI/CD.

- Built reproducible ML training pipelines using **MLflow** and **DVC**, integrating model versioning with data lineage tracking for audit-compliant deployments.
- Containerized model training and inference workflows using **Docker**, ensuring consistent execution across development, staging, and production.
- Automated model training and validation pipelines on **AWS SageMaker**, enabling parallel hyperparameter tuning and GPU-based acceleration for deep learning models.
- Designed Helm-based deployment templates on **Kubernetes** for serving MLflow-tracked models as scalable REST endpoints with auto-scaling and health probes.
- Integrated **GitHub Actions** and **Jenkins** to orchestrate end-to-end CI/CD workflows for data preprocessing, model training, testing, and deployment.
- Developed custom **REST APIs** in **Python** to abstract experiment lifecycle actions (register, promote, deploy) for external orchestration tools.
- Enforced code and model quality using **PyTest**, **Flake8**, and **pre-commit hooks**, ensuring adherence to internal ML code compliance guidelines.
- Implemented experiment tracking dashboards with **MLflow UI** and custom metadata logging for performance benchmarking across model versions.
- Automated retraining triggers based on data drift detection metrics and scheduled workflows via **GitHub Actions** runners.

- Designed namespace-isolated environments in **Kubernetes** to safely run multi-tenant experimentation without resource contention or cross-contamination.
- Coordinated with data scientists to refactor research notebooks into production-ready **Jupyter Notebook** pipelines, decoupled from UI and infrastructure concerns.
- Participated in **Agile** ceremonies and bi-weekly demos, actively translating platform feedback into actionable enhancements on version control, rollout stability, and traceability.

Environment: Python, MLflow, DVC, Docker, Kubernetes, AWS SageMaker, GitHub Actions, Jenkins, REST APIs, CI/CD, Jupyter Notebook, PyTest, Flake8, Agile

HighRadius (Fiserv) | New York, US – Bengaluru, India | Data Engineer

Oct 2018 – May 2021

- Built credit risk models using **XGBoost** and **LightGBM** to score invoice default probabilities, deployed via **Flask** APIs.
- Engineered features with **Pandas** and **NumPy**, applying **SHAP** for model interpretability in payment behavior models.
- Developed **spaCy** and **NLTK** pipelines to extract invoice fields from remittance texts, automating reconciliation tasks.
- Used **Statsmodels** for regression analysis on customer aging data to support collections prioritization.
- Deployed models on **AWS EC2** in **Docker** containers with health checks and rollback safety for production scoring.
- Processed historical transactions from **AWS S3** for time series modeling and anomaly detection.
- Created risk heatmaps and payment trend plots using **Plotly** and **Seaborn** for collections teams.
- Tuned models with **scikit-learn** cross-validation and early stopping for high-cardinality invoice datasets.
- Queried and joined data from **PostgreSQL** and **MySQL** to create unified training datasets.
- Versioned code and ML experiments via **Git**, maintaining reproducibility across model iterations.
- Exposed scoring endpoints via **Flask**, with token-based auth and test coverage for SLA compliance.

Environment: Python, scikit-learn, XGBoost, LightGBM, Statsmodels, Pandas, NumPy, Flask, Docker, AWS EC2, AWS S3, spaCy, NLTK, Plotly, Seaborn, Git, PostgreSQL, MySQL

Simform | Bengaluru, India | Data Analyst

May 2016 – Sept 2018

- Built ingestion scripts using **Python**, **Pandas**, and **SQLAlchemy** to pull and normalize transactional data from **MySQL** and **PostgreSQL**, improving pipeline runtime.
- Trained and evaluated churn prediction models using **scikit-learn**, enabling targeted retention actions for high-risk users.
- Applied statistical tests in **R** to analyze A/B test outcomes for product and marketing changes, driving data-backed deployment decisions.
- Developed **ARIMA** and **Prophet** models via **Statsmodels** to forecast weekly sales trends, improving inventory planning accuracy.
- Cleaned and transformed large datasets using **NumPy** and **Pandas**, including outlier filtering, scaling, and encoding, to prepare features for model input.
- Built Tableau dashboards integrating model outputs and KPIs to support data access for marketing and sales teams.
- Automated exploratory data analysis in **Jupyter** using custom **Python** templates for column profiling, correlation checks, and null handling.
- Used **Git** for version control and collaboration, managing code branching and change tracking in cross-functional development workflows.
- Verified model reproducibility in **Ubuntu (WSL)** environments to align with deployment targets and reduce post-handoff issues.

Environment – Python, R, Pandas, NumPy, scikit-learn, Statsmodels, MySQL, PostgreSQL, Tableau, Jupyter, VS Code, Git, Ubuntu, WSL